Act Through the Latent: Closing the Retarget-to-Track Interface with a Goal-Conditioned Command Prior

Anonymous ICRA Submission

***Abstract*—The dominant humanoid motion pipeline is two-stage: retarget human motion to the robot, then train a tracking policy on the result. The interface between the stages is a stack of robot-space joint targets—a representation chosen for convenience, not for control. We take a first step toward a *learned* interface. We distill a per-frame IK retargeter into one multi-embodiment network with a shared retargeting latent z_{ret} (128-d, five humanoids), and audit that latent with pre-specified probes: it is instance-aligned across embodiments (window retrieval 75–83% at 1.02% chance; CKA 0.91) but carries little semantic motion category (0.151 vs 0.125 chance), is embodiment-aligned but not invariant (0.28 linear probe vs 0.91 nonlinear attacker), physically impoverished (contact F1 0.088 without supervision), and—frozen—inert as a control input, a null that an independent ablation confirms. On the control side we show a separate 64-d command bottleneck z_{cmd}, produced by a goal-conditioned residual-to-prior CVAE and DAgger-distilled from an explicit-command RL teacher, can be the actor's *only* motion input at 0.952 ± 0.002 ten-second completion versus the 0.98 teacher (three seeds), where naive alternatives fail (frozen concatenation is harmful; goal-free priors reach 0.02–0.05). The retargeting network also absorbs a second, interaction-rich teacher across three human skeletons, cutting interaction-clip articulation error 3–$5\times$ at $+1.05\,$cm on locomotion (preliminary: ground-truth root). We release a pre-specified negative-results ledger and a measurement-substrate defect whose repair cut joint tracking error by 46%.**

I. Introduction

Nearly every humanoid that has learned to move from human motion has done so in two stages: a retargeter converts human motion into robot-space joint trajectories, and a reinforcement-learning tracking policy learns to follow them. The split is a good engineering decision—retargeters and trackers are developed, evaluated, and reused independently, and the recent generation of each is strong: interaction-preserving optimization on the retargeting side [2], physics-in-the-loop retargeting [1], and unified tracking controllers that follow thousands of clips with one network [4], [5]. But the *interface* between the stages has escaped scrutiny. What actually crosses the boundary is a per-frame stack of joint targets—a format, not a representation.

That format is lossy in three specific ways. It discards everything the retargeter knew that is not a joint angle: its contact reasoning, its uncertainty, and the cross-embodiment structure it learned when one network serves several robots. It is evaluated by kinematic metrics (pose error, foot skate, penetration) that are blind to *trackability*—the GMR study [3] showed retargeting artifacts silently cap what downstream RL can learn, evidence that the interface transmits harm as readily as signal. And it is *dimensionally unshareable*: a 29-DoF joint-target stream cannot command a 23-DoF robot, so every multi-robot system pays the interface cost once per embodiment.

The obvious repair fails, and the failure is informative. Handing the tracking policy the retargeter's latent *alongside* its usual reference does nothing: in our experiments the policy ignores the latent (concatenation is null-to-harmful, -10 points at worst), and UniTracker's ablation reports the same mechanism independently—"when the actor receives the reference motion directly, the influence of the latent variable z vanishes" [6]. A frozen latent as the *sole* command is feasible but costs about thirty points against the explicit-command teacher under the identical training recipe (0.65 vs 0.98). Training a latent by RL from scratch collapses [6], [15]. The latent, as produced by retargeting alone, is not a control interface.

Our key insight is that the retargeting-to-tracking boundary should not be a representation the two stages agree on, but one the tracking policy learns to act through: a latent becomes useful for control exactly when it is the policy's only channel to the goal, and is inert whenever an explicit reference is available beside it. The sections that follow are the constructive and destructive halves of that claim. We build SNMR, a skeleton-agnostic retargeting network distilled from a classical IK teacher whose shared latent serves five humanoids, and we audit that latent with falsification-first probes before asking it to do anything (§IV). We then make the interface load-bearing (§V): a conditional VAE whose goal-conditioned prior compresses the reference into a 64-d command latent, trained by DAgger from an explicit-command RL teacher, with the reference visible to the prior but never to the decoder. On a simulated Unitree G1 this bottleneck reaches 0.952 ± 0.002 ten-second completion against its 0.98 teacher across three seeds (Fig. 1c); its prior still consumes the robot-space goal, so we present it as the actor-side half of the interface program, with the source-side prior specified as the open step. The same latent then absorbs a second, interaction-rich teacher—terrain and object clips from a fundamentally different retargeter, arriving on different human skeletons—cutting interaction-clip error 3–$5\times$ at a measured $+1.05\,$cm locomotion cost (§VI).

Two findings we did not go looking for became contributions. First, a world-body indexing defect in a widely used MuJoCo-Warp training framework silently zeroed the

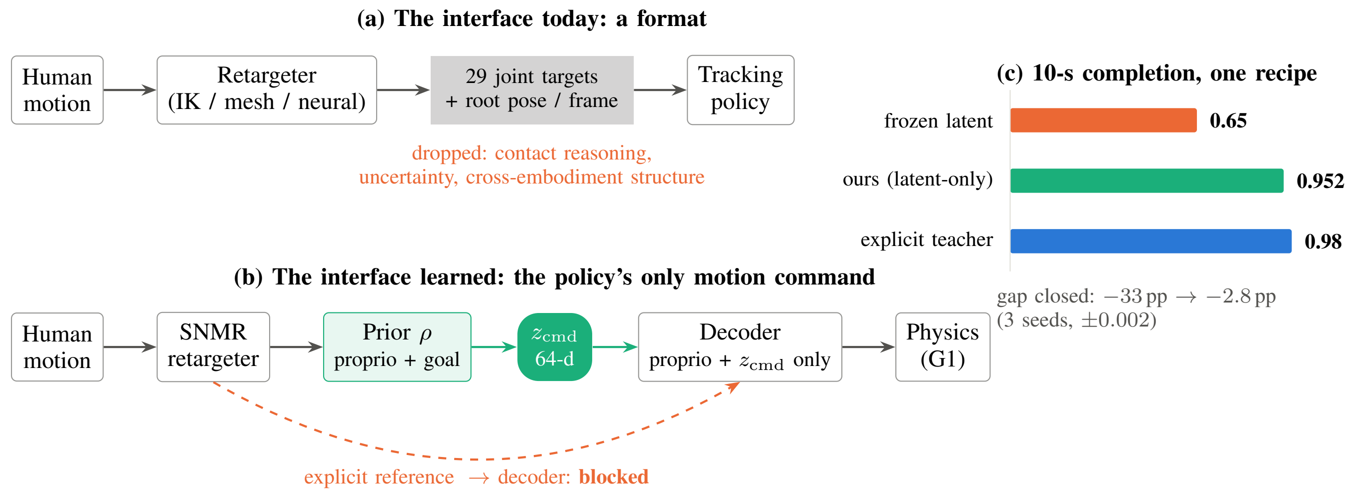

Fig. 1. **The retarget-to-track interface, replaced.** (a) The standard humanoid stack passes per-frame joint targets between retargeting and tracking, discarding the retargeter's contact reasoning, its uncertainty, and the cross-embodiment structure it learned. (b) We replace that interface with a learned 64-d command latent: the reference reaches the goal-conditioned prior but *never* the decoder, so z_{cmd} is the tracking policy's only motion command. (c) Under one training regime, the frozen retargeting latent reaches 0.65 ten-second completion, our co-trained command latent 0.952 ± 0.002, and the explicit-command teacher 0.98: co-training closes the interface gap from thirty-three points to under three.

primary body-position tracking reward in every one of our runs (upstream report in preparation); its repair cut joint tracking RMSE by 46% with no change to the learning algorithm and brought our tracker into an externally calibrated band (§VII). Second, rolling tracking policies out on their own references and recording the simulated states yields physically repaired supervision—2–3× less foot skate at a measured fidelity price—quantifying the data-side alternative to physics-in-the-loop retargeting.

Contributions.

- **A learned command bottleneck for humanoid tracking.** A goal-conditioned, residual-to-prior CVAE whose 64-d latent is the policy's only motion input—0.952 ± 0.002 completion against a 0.98 explicit-command teacher (3 training seeds, SD), with the reference never visible to the decoder, and a documented failure anatomy (path consistency, goal conditioning) for making such bottlenecks work.
- **A falsification-first audit of a shared motion latent.** Preregistered probe families establishing what the latent carries—content-rich, embodiment-aligned but not invariant, physically impoverished, control-inert while frozen—with the negative verdicts stated as measured results, one independently replicated.
- **Skeleton-agnostic retargeting distillation, measured honestly.** A 1.5M-parameter network amortizing an IK teacher across five humanoids (limits by construction, 671 fps CPU), and a separate two-teacher G1 study absorbing interaction-rich clips across three human skeletons at a quantified locomotion cost (preliminary: articulation under ground-truth root)—with the sharing cost and the zero-shot-embodiment failure (5.2×) measured rather than assumed.
- **A measurement-substrate defect, found, fixed, and released**, plus the pre-specified negative-results ledger—closed research lines with the evidence that closed them, so the field can stop re-deriving them.

II. Related Work

Retargeting as preprocessing. Classical retargeting solves per-frame IK with hand-tuned correspondences [3]; interaction-mesh methods add hard contact constraints [2]; ReActor moves retargeting inside physics via bilevel RL, eliminating artifacts by construction [1]. All three families emit the same interface—robot joint trajectories—and are evaluated by kinematic quality plus, recently, downstream RL success [3], [2]. None asks whether the *representation* handed to the tracker should itself be learned. Neural feed-forward retargeters [12], [13], [14] amortize the optimization and, in AdaMorph's case, span many embodiments through a morphology-agnostic latent—but their latents are internal machinery, not command interfaces, and none is analyzed for what it carries.

Tracking and its command spaces. Modern trackers follow references specified as joint targets with body-pose terms [4], [5], [11]; HOVER unifies multiple command *modes* by distillation [10]; UniTracker compresses the reference into a CVAE latent and is the closest precedent for our Stage-2—single-embodiment, with no retargeter and no representation analysis [6]. Latent skill spaces for physics characters [7], [8], [9] establish the recipe elements we verify at robot scale: learned conditional priors, residual-to-prior encoders, distillation-not-RL. Behavioral foundation models [16] reach a promptable latent from the opposite entry point—unsupervised RL with no retargeting anywhere. To our knowledge no published system co-trains a *multi-embodiment retargeting* latent under the *control* objective.

Interface critiques. BeyondMimic names a "planning-control gap" [4]; the GMR study quantifies how retargeting quality caps tracking [3]. Both stop at the boundary we cross: neither changes what the boundary *is*.

III. The Two-Stage Interface

A source motion m_t enters a retargeter R; the standard interface hands the tracker π a reference $g_t = (q_t^{\mathrm{ref}}, \dot{q}_t^{\mathrm{ref}})$ of robot joint targets. We move the actor-visible reference behind a learned bottleneck: $z_{\mathrm{cmd}} = \mu_\rho(\mathrm{proprio}_t, g_t)$ (the current-frame command; a future window is future work), under an *exclusivity contract*: the goal is visible to the prior ρ and to the critic, but never to the decoder, so motion intent reaches the actor only through the 64-d bottleneck. We emphasize scope: the prior currently consumes the *robot-space* goal g_t, so this is an actor-side bottleneck within the tracker; the interface-replacement step—a prior over the source-side representation z_{ret} alone—is specified in §VIII.

The latent's source is SNMR: a graph-attention encoder over the source skeleton (per-node features, adjacency, global pooling—skeleton-agnostic by construction) maps motion to a per-frame retargeting latent $z_{\mathrm{ret}} \in \mathbb{R}^{128}$ (distinct from the control bottleneck z_{cmd}); embodiment-conditioned graph decoders (AdaLN on a 32-d embodiment code, $\tanh$ joint-limit heads) emit trajectories for any of five humanoids. Trained by distillation from a per-frame IK teacher [3] on 77 LAFAN1 clips $\times$ 5 robots (2.48M paired frames), it reaches 3.66 cm held-out MPJPE (CI $[3.46, 3.86]$; G1 specialist; 2.9–6.0 cm shared across all five robots) at 671 fps on CPU, with zero joint-limit violations by construction. Foot skate (0.29 vs teacher 0.05 m/s) is the known inherited gap and is treated as a measured limitation, not hidden.

IV. Auditing the Latent

Before asking the latent to carry control, we measured what it holds. Four pre-specified probe families, in increasing severity:

Content: instance-aligned, not semantic. Cross-embodiment *window* retrieval hits 75–83% top-1 at 1.02% chance (98 windows); CKA between per-robot encodings is 0.89–0.92. But a motion-*category* probe is near chance (0.151 vs 0.125): the latent aligns instances across bodies without organizing them semantically.

Embodiment: aligned, not invariant. A linear probe recovers the source embodiment at only 0.28 accuracy, but a nonlinear attacker reaches 0.91—the classic gap between "not linearly decodable" and "absent." Scale explains only $\sim$1% of the leak.

Physics: impoverished. Foot-contact state is weakly present under pure distillation (linear F1 0.088 at full budget). A co-trained contact head injects it durably (F1 0.227 at full budget, $1.8\times$ a deployable-heuristic baseline) at a real fidelity price (+1.5 cm at loss weight 0.5), so physics enters the latent only when supervised in.

Control: inert while frozen. Concatenated beside the explicit reference, the latent is ignored or harmful (-10 pp); as the sole command it reaches 0.65 under our final recipe; a learned readout over the frozen latent does not close the gap. The audit is the specification for §V: nothing short of co-training makes this representation an interface.

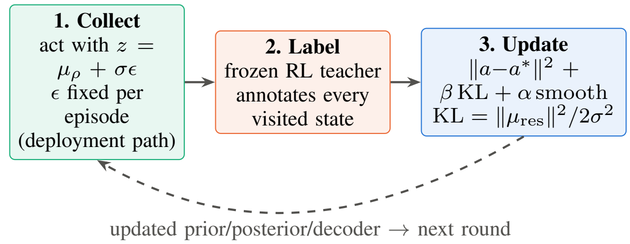

Fig. 2. **One DAgger round (×2,000).** Rollouts are collected on the deployment (prior) path with episodic noise; a frozen explicit-command teacher labels visited states; the CVAE updates with action matching, a closed-form KL from the residual-to-prior posterior (tied σ), and a latent smoothness term. The dashed return edge makes it DAgger rather than behavior cloning. At deployment: $z_{\mathrm{cmd}} = \mu_\rho$ (deterministic), no teacher, no posterior, no explicit reference to the actor.

V. Making the Latent Load-Bearing

The method (Fig. 2) is a conditional VAE trained by online distillation. The **prior** $\rho(z_{\mathrm{cmd}} \mid \mathrm{proprio}, g)$ sees deployable inputs only: proprioception plus the reference goal window (and, in one arm, the frozen SNMR latent stream). The **posterior** adds privileged simulation state as a *residual on the prior mean* with tied fixed σ=0.3, giving the closed-form KL $\|\mu_{\mathrm{res}}\|^2/2\sigma^2$. The **decoder** maps $(\mathrm{proprio}, z_{\mathrm{cmd}})$ to PD targets at 50 Hz and never sees the goal. Training is pure DAgger from a frozen explicit-command RL teacher (no RL gradient mixed in), with episodic reparameterization noise and a latent smoothness regularizer.

Three failures fixed the design, in order. (i) Collecting roll-outs on the privileged posterior path collapses deployment to 0.001 while the posterior path itself scores 0.84—train/deploy path consistency is not optional. (ii) Fixing the collection path alone recovers only 20×; mixing prior-path samples into the action loss damages the posterior (0.84 → 0.42). (iii) Adding goal conditioning to the prior while removing prior-path samples from the action loss lifts the deployment path to 0.935–0.955; a factorial ablation separating those two changes is in progress, so we report the combination rather than a single-factor attribution. The working design compresses to one line: *the prior sees the goal; the posterior adds only a small privileged residual; the decoder sees neither.*

VI. Experiments

All tracking numbers: 10-second phase-stratified rollouts on a simulated Unitree G1 (MuJoCo-Warp), deterministic deployment path, 1,024 rollouts per evaluation for latent arms (100 for reference policies). Retargeting numbers: 7 held-out LAFAN1 clips, bootstrap CIs over per-clip means.

A. Does the learned interface work?

Table I is the paper's central result. A 64-d co-trained bottleneck as the actor's only motion input recovers 97% of

TABLE I
COMMAND-INTERFACE COMPARISON, ONE TRAINING REGIME (G1, WALK1).

Interface (actor's motion input)	Completion	RMSE (rad)	Seeds
Explicit command (teacher)	0.980	0.122	1
Ours: latent-only, +SNMR z	$\mathbf{0.952 \pm 0.002}$	0.127	3
Ours: latent-only	0.944 ± 0.008	0.128	3
Frozen SNMR latent	0.650	—	1

TABLE II
TWO-TEACHER ABSORPTION, *preliminary*: HELD-OUT ARTICULATION MPJPE (CM) UNDER GROUND-TRUTH ROOT POSE; MATCHED STEP BUDGET.

Held-out pool	LAFAN1-only	Two-teacher	Δ
Locomotion (LAFAN1)	**3.31**	4.36	$+1.05$
Object interaction	28.95	**5.53**	-23.4 (5.2×)
Terrain interaction	25.06	**8.16**	-16.9 (3.1×)

its explicit-command teacher's completion at matched joint RMSE (same DAgger algorithm, same reward stack, three seeds). The frozen retargeting latent as sole command reaches 0.65 under a PPO recipe; because algorithm, interaction budget, and seed count differ from the student runs, we report it as context rather than a controlled contrast—the within-recipe frozen baseline (identical DAgger pipeline, z_{ret} command) is running and will replace this row. What Table I does establish: the bottleneck costs only 2.8 points against its own teacher, where the design's failed variants (posterior-path collection, goal-free priors) collapse to 0.001–0.05.

B. Does the latent add value beyond the explicit goal?

On single-clip walking: no—the paired difference between the +SNMR-latent arm and the goal-only arm straddles zero across seeds ($+2.1/+0.5/-0.2$ pp), exactly as pre-specified (an explicit goal saturates a single clip). The +SNMR arm's seed variance is 4× tighter (±0.002 vs ±0.008; suggestive at n=3). Multi-clip teachers at our compute budget have not yet cleared their quality gate (0.47 mean completion at 1024 envs × 16k iters over 8 heterogeneous clips), so the multi-clip and cross-embodiment answers—where explicit commands are dimensionally unshareable and the latent's value is structural—are reported as in-progress rather than claimed.

C. Does the interface scale with better retargeting data?

Preliminarily, yes (Table II; articulation under ground-truth root—object/terrain state is not yet input, and a redo with full root loss and group-held-out splits is registered). Adding a second, interaction-rich teacher—1,938 OmniRetarget clips [2] whose paired human data arrives on 52- and 53-joint keypoint skeletons, ingested with MST-inferred topology and synthetic orientations—cuts held-out interaction-clip error 5.2× (object) and 3.1× (terrain) while locomotion pays $+1.05$ cm. A sampling-diet ablation attributes 61% of the naive run's larger interference to data balance rather than representational conflict. The terrain plateau (∼8.2 cm vs 5.5 for object) is the measured signature of a hidden conditioning variable (terrain scale): the dataset contains 4,659 sibling pairs with *bit-identical* human motion and divergent robot trajectories (mean DoF divergence 0.031–0.068 rad)—real conditional multimodality that a deterministic decoder must average, and that we quantify at 16–36× the threshold below which our earlier work showed generative decoders have nothing to model.

D. Are learned references as trackable as IK references?

Matched tracking policies (3 seeds per source × 2 eval seeds × 100 rollouts) trained on SNMR references vs the IK teacher's: 86.7% vs 86.7% completion—point-equal—with joint RMSE noninferior (+0.33%, CI $[-3.1, +4.9]$%). This matrix predates the reward repair (§VII); both sources trained under the identical defect, but a position-blind assay may under-detect source differences, so a post-repair rerun with a sensitivity control (deliberately corrupted references) is registered and pending. Formal noninferiority on completion was not established: the ±6 pp CI exceeds the pre-specified -5 pp margin at this seed count—a power statement, not a quality gap (per-seed spread ±3 pp dominates any source effect).

E. What does physics-repaired supervision buy and cost?

Rolling a calibrated tracker out on its own references and recording simulated states yields repaired supervision with 2–3× lower stance-foot speed than the references themselves (3–6× on pre-repair policies) and zero penetration (a simulator property, which we do not claim), at 6.2–6.7 cm heading-local reconstruction distance—the quantified data-side alternative to bilevel physics retargeting [1].

VII. A DEFECT IN THE MEASUREMENT SUBSTRATE

While calibrating our tracker against external baselines we found that the primary body-position tracking reward had been *exactly zero* for every one of our MuJoCo-Warp runs: a world-body off-by-one (state tensors include the world body at index 0; the body-name list excludes it) shifted every simulation-side body read one body rootward in our stack (pinned revision; IsaacSim/IsaacGym paths unaffected), so the tracked pelvis slot read the world body—always at the origin—and the exponential reward kernel underflowed to 0.0 for 8,000 consecutive iterations (the raw term logs prove it). Policies still walked, because orientation and velocity terms survived; they were orientation-plus-velocity trackers with a corrupted position channel. Repairing the indexing (released as a small patch; IsaacSim/IsaacGym paths are unaffected, which is why upstream never saw it) cut joint tracking RMSE from 0.263 to 0.142 rad (−46%) with no change to the learning algorithm; adding a standard joint-space reward term then reached 0.122 rad at 0.98 completion, matching an external 187-run calibration band (97.9%). Paired comparisons made under the defect survive (both sources suffered identically); absolute capability numbers did not. We report this at section level because the lesson generalizes: *measure your reward channels before tuning them.*

VIII. Limitations

Interface scope — the central open step. The current prior consumes the robot-space goal; the interface-replacement experiment (a prior over z_{ret} alone, with tracking gradients optionally reaching the retargeting encoder) is specified and queued, as are the within-recipe frozen baseline and the factorial attribution ablation. The latent's additive value over an explicit goal is null on single-clip walking; multi-clip and cross-embodiment settings are in progress. **Embodiment generalization.** Zero-shot transfer to an unseen robot fails ($5.2\times$ error); embodiment augmentation is the identified path. **Physics of the latent.** Contact enters only by co-training at a fidelity price; repaired references cost 6+ cm at current tracker quality. **Interaction tracking.** Our simulator path supports neither objects nor terrain in the tracking loop; interaction data enters at the retargeting level only, where its absorption is measured. **Simulation only.** No hardware; the MuJoCo-Warp stack is externally calibrated but sim-to-real is unvalidated. **Data scale.** 4.6 h/robot of locomotion plus 0.9–3.4 h of interaction data is small; the two-teacher result is the first point on the scaling curve, not its end. Moving past these—conditioning on the interaction hidden variable, capacity scaling matched to data, and the cross-embodiment command demonstration—is the immediate agenda.

IX. Conclusion

The boundary between retargeting and tracking has been a file format. Treated as a representation—audited before trusted, co-trained before deployed, and made the policy's only channel to the goal—it carries nearly everything the explicit interface carried, in a form one network can share across robots and extend across teachers. The measurement lesson runs deeper than the architecture: two of this paper's strongest results are a reward channel that was silently dead and a set of pre-specified nulls; we commend both practices to the field.

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